

Classical baselines for multi-rotor RPS estimation

Why they fail and by how much

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1 Classical methods we tried

The task is to recover four per-rotor speeds from a single-channel mixture of speech + drone noise at very low SNR (-30 to 0 dB). Before training a neural network it is worth checking whether standard signal-processing methods can do the job. We picked four well-known single-pitch estimators and extended each to the multi-rotor case with the same greedy suppression procedure.

PYIN. The probabilistic YIN tracker from `librosa`. It is a single- f_0 method; we run it with a wide frequency range $[50, 400]$ Hz to capture both the motor rotation frequency and the blade-passage frequency, then divide by the blade count ($b = 2$) when necessary. Because PYIN can only track one source, we replicate its single output for all four rotors.

Cepstral analysis. For each STFT frame we compute the real cepstrum $c = \mathcal{F}^{-1}\{\log |\mathcal{F}\{\mathbf{s}\}|\}$ and search for the strongest peak in the quefrency band that maps to the plausible RPS range $[50, 150]$ rev/s. The quefrency peak is converted back to frequency via $f = f_s/q$ and then to rotor speed $\rho = f/b$.

Harmonic Product Spectrum (HPS). The spectrum is multiplied with downsampled copies of itself:

$$P_{\text{HPS}}[k] = |S[k]| \cdot |S[2k]| \cdot |S[3k]| \cdot |S[4k]|. \quad (1)$$

The peak of P_{HPS} in the band $[b\rho_{\min}, b\rho_{\max}]$ gives the blade-passage frequency; dividing by b yields the rotor speed.

Matched-filter bank. We build a dense grid of candidate rotor speeds $\rho \in \{50, 50.5, \dots, 150\}$ rev/s and, for each candidate, a binary comb template with narrow windows around its first 15 harmonics. The template that maximises the cosine similarity with the magnitude spectrum is taken as the estimate.

NMF with harmonic dictionary. Semi-supervised non-negative matrix factorisation: the magnitude spectrogram $\mathbf{V}$ is factorised as $\mathbf{V} \approx \mathbf{W}\mathbf{H}$ where $\mathbf{W}$ is a fixed dictionary of Gaussian harmonic-comb spectral templates (one column per candidate RPS, same grid as the matched filter). $\mathbf{H}$ is learned via multiplicative updates; for each frame the four strongest activations are taken as the rotor-speed estimates. This is a true multi-pitch method—there is no greedy suppression step.

All methods share the same constants: $f_s = 16$ kHz, $N = 2048$, $H = 512$, blade count $b = 2$, and 15 harmonics.

2 The greedy multi-rotor extension

The first four methods estimate *one* rotor speed per frame. To get four estimates we use the same greedy suppression scheme for all of them:

Algorithm 1 Greedy multi-rotor extractor

Require: Frame spectra $\mathbf{S} \in \mathbb{R}^{T \times F}$, single-rotor estimator $\mathcal{E}$, rotors $R = 4$, suppression half-width $w = 3$ bins

Ensure: $\hat{\mathbf{R}} \in \mathbb{R}^{R \times T}$

```

1: for  $t = 1 \dots T$  do
2:    $\mathbf{s} \leftarrow \mathbf{S}[t, :]$  ▷ copy
3:   for  $r = 1 \dots R$  do
4:      $\hat{\rho}_{r,t} \leftarrow \mathcal{E}(\mathbf{s})$ 
5:     for  $h = 1 \dots 15$  do
6:        $k_h \leftarrow \text{round}(h b \hat{\rho}_{r,t} N / f_s)$ 
7:        $\mathbf{s}[k_h - w \dots k_h + w] \leftarrow 0$ 
8:     end for
9:   end for
10: end for
```

This is simple, deterministic, and wrong in several ways. Hard zeroing removes not only the target rotor but also overlapping speech energy and sidebands from other rotors, so the spectrum gets progressively more damaged for the second, third and fourth rotors. Each frame is processed independently, so there is no temporal continuity. And the first rotor always gets the cleanest spectrum while the others fight for scraps. These are not implementation bugs; they are structural limitations that any single-pitch method extended this way will hit.

3 Learned baseline for comparison

SimpleConv is a small CNN (≈ 1 M parameters) that maps the log-magnitude STFT of the mixture directly to four per-rotor speed traces via frame-wise regression. It has no explicit harmonic model, no suppression step, and no hand-designed peak picking. The encoder is five 2-D conv blocks that downsample frequency while keeping time intact; a spatial mean pools the remaining frequency bins and a two-layer 1-D conv head outputs four channels. It is trained end-to-end on DREGON-LM by MSE against telemetry-aligned ground truth. See the architecture table in the main paper if you care about the exact layer dimensions; for this report the only thing that matters is that it is a plain neural network that learns the mapping from data.

4 Results

4.1 Numbers

Table 1 shows mean and standard deviation across the 10 DREGON-LM test samples. The classical methods are not just worse—they are *two to three orders of magnitude* worse.

NMF is the best classical method (MSE 615), roughly $86\times$ worse than SimpleConv but clearly ahead of the greedy single-pitch methods (870–1380). Cepstral and matched-filter are $193\times$ and $187\times$ worse. The R^2 values are deep negative, which means even the best classical predictions are worse than predicting the per-sample mean. In other words, these methods do not just fail to track the rotors; they actively make things worse.

Table 1: Aggregate metrics on 10 test samples.

method	mse		mae		r2	
	mean	std	mean	std	mean	std
simple_conv	7.16	11.34	1.70	1.06	-1.20	3.65
pyin	623.11	608.95	22.08	8.22	-229.45	154.81
cepstral	1379.03	514.57	28.81	4.89	-600.26	362.57
hps	870.46	152.71	27.56	1.47	-438.82	255.23
matched_filter	1338.59	301.77	29.79	2.92	-666.20	384.96
nmf	614.56	443.66	18.91	5.17	-244.56	147.91

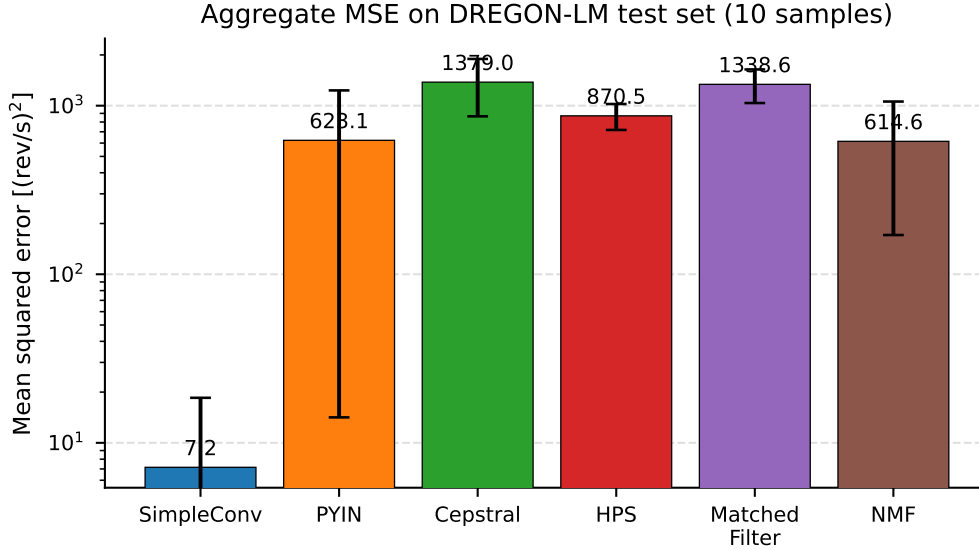


Figure 1: Mean MSE on a log scale. Error bars are ± 1 std.

Table 2 breaks it down per sample. SimpleConv stays below 40 on every single clip. The classical methods never drop below 160 and routinely exceed 1000.

4.2 What the predictions actually look like

Mean RPS is misleading because the four rotors move independently and the mean can be flat even when individual rotors are wrong. Figures 2 to 4 show the per-rotor predictions for individual samples. Each page is one sample: spectrogram on top, then six per-rotor panels (one per method). Dotted lines are ground truth; solid lines are predictions. Rotor colours are consistent across all panels.

A few things are immediately obvious from the plots:

- **SimpleConv** actually follows the four dotted traces. It is not perfect—there is some overshoot and occasional mixing of rotors—but the solid lines are clearly correlated with the ground truth.
- **PYIN** returns a single trace replicated four times. When it finds voiced frames it locks onto whatever is strongest in the mixture, which may be speech or a single rotor, and ignores the other three.
- **Cepstral**, **HPS**, and **Matched Filter** were at least extended with greedy suppression, but the second, third and fourth estimates are driven by random spectral peaks because

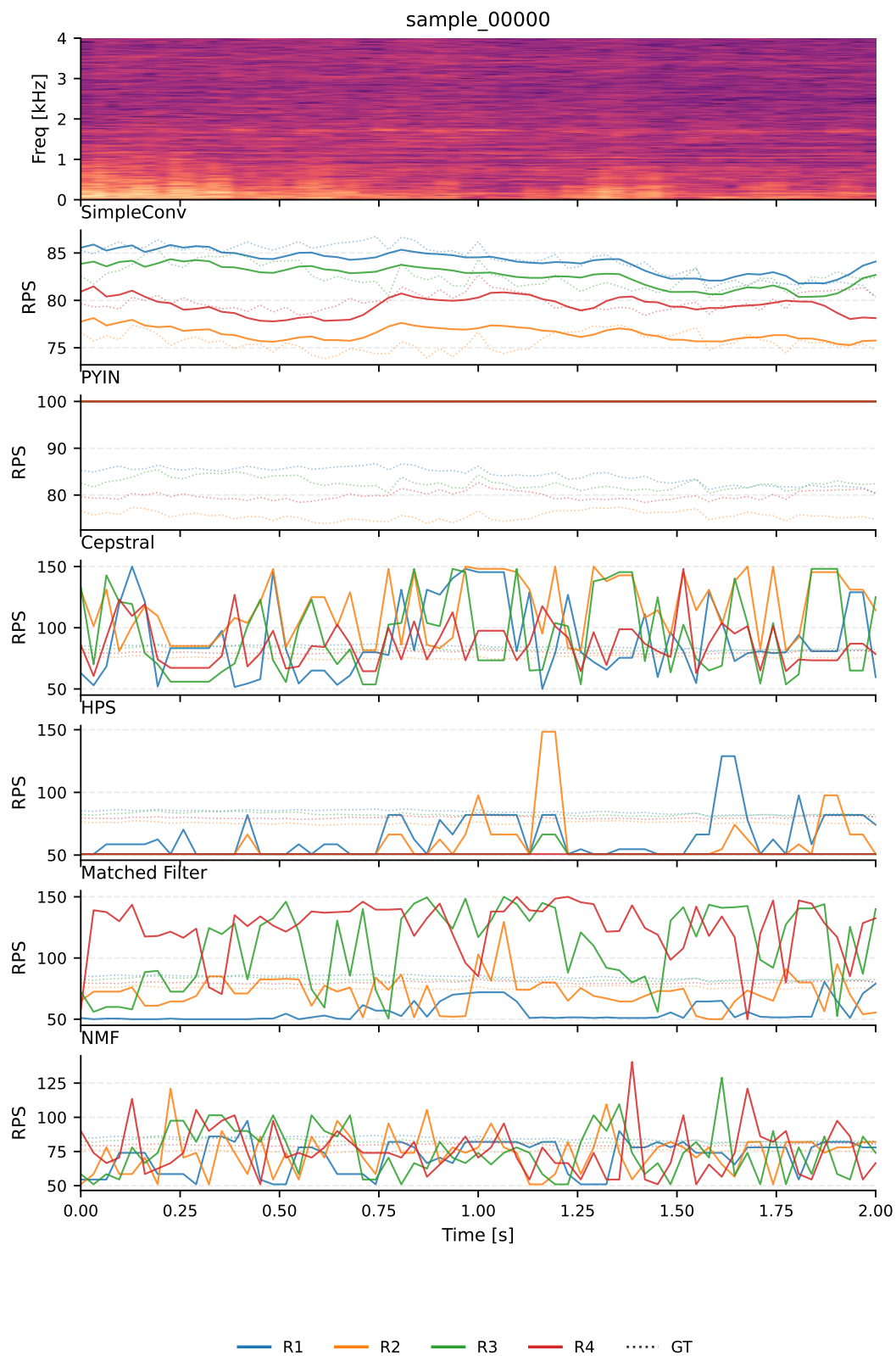


Figure 2: Sample 00000 – the easy case. SimpleConv tracks all four rotors closely; the classical methods are essentially flat or random.

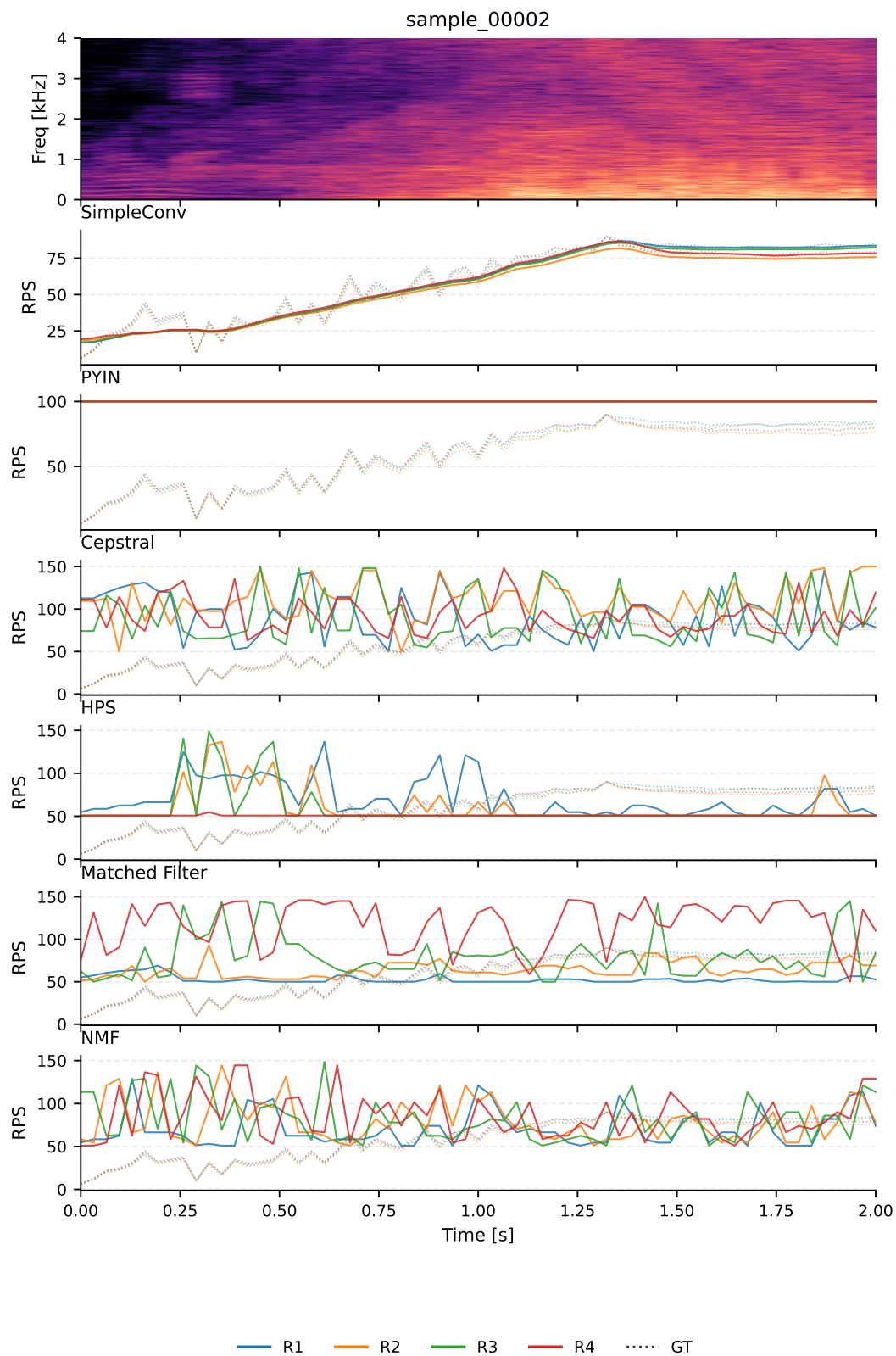


Figure 3: Sample 00002 – a harder clip where SimpleConv MSE rises to 38 but still remains two orders of magnitude below the classical methods.

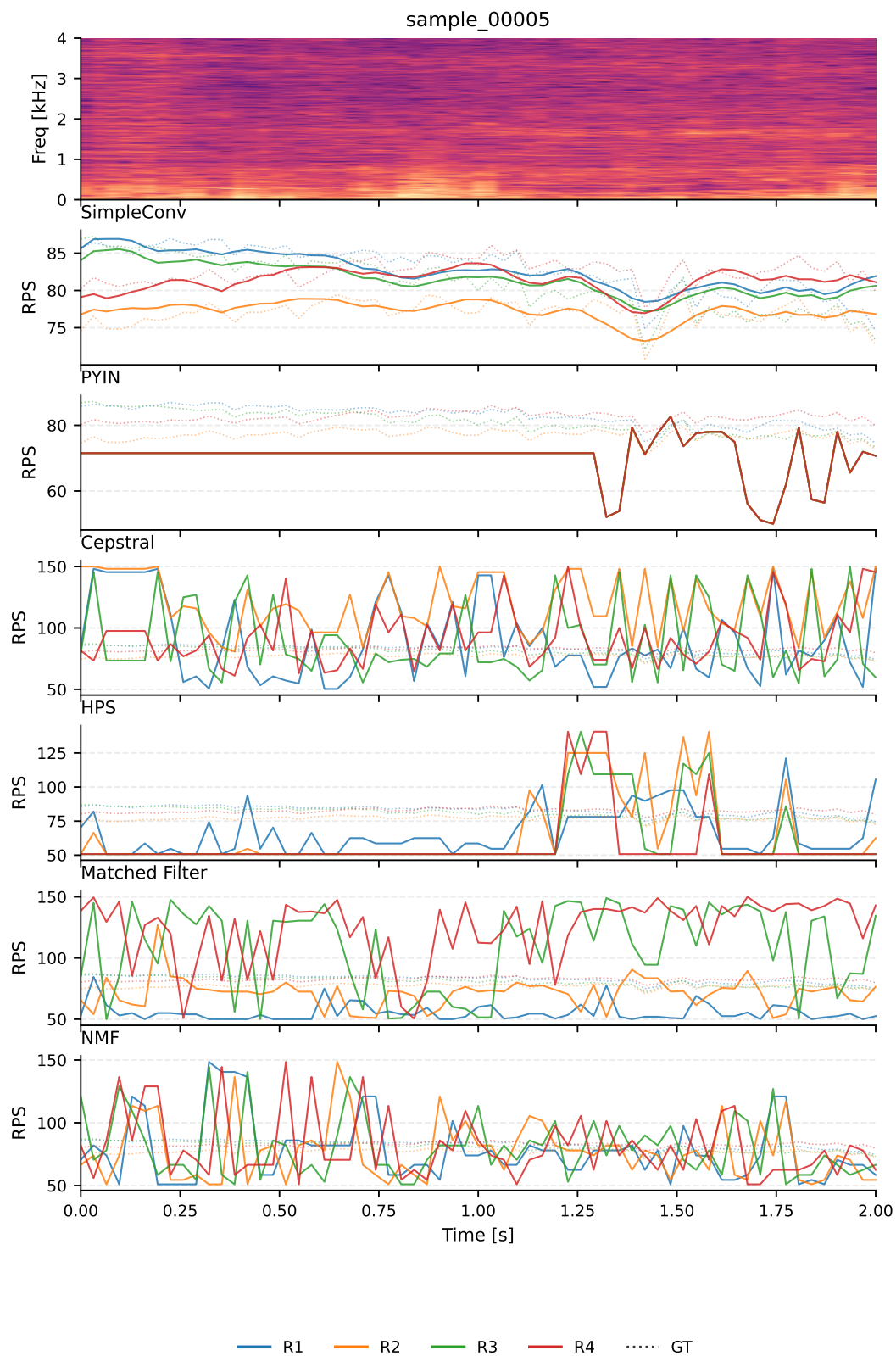


Figure 4: Sample 00005 – one of the few clips where PYIN finds voiced frames. It still tracks only one periodicity and replicates it for all four rotors.

Table 2: Per-sample MSE [(rev/s)²].

method sample	simple_conv	pyin	cepstral	hps	matched_filter	nmf
sample_00000	1.4	389.6	1226.8	796.3	1336.6	316.0
sample_00001	2.8	362.3	1315.3	867.8	1227.4	407.7
sample_00002	38.0	2270.4	2819.7	1289.2	1935.0	1818.2
sample_00003	2.1	886.7	1197.8	805.6	1194.4	451.7
sample_00004	1.2	390.4	1231.9	847.5	1075.7	523.6
sample_00005	3.9	161.2	1278.5	889.0	1344.3	594.6
sample_00006	2.0	600.1	1144.8	840.6	1120.4	410.0
sample_00007	1.9	366.9	1000.0	823.0	1017.2	796.4
sample_00008	6.0	398.3	1300.8	802.5	1337.4	406.2
sample_00009	12.3	405.2	1274.7	743.1	1797.6	421.2

hard bin zeroing is too destructive.

- **NMF** (bottom panel) does better than the greedy methods: its traces sometimes follow the correct rotor for a few frames before losing track. But it still drifts and confuses rotors when harmonics overlap.

5 Why they fail

The main reason is embarrassingly obvious: four of the five methods are not multi-pitch estimators.

Cepstral analysis, HPS, and the matched-filter bank are all single-pitch algorithms. They are designed to find one dominant periodicity in a spectrum. To make them handle four rotors we added a greedy extension that finds the strongest pitch, suppresses its harmonics by zeroing out bins, and repeats. This is not how multi-pitch estimation works in the literature; it is a crude hack that we tried because it was easy to implement. The problem is that zeroing a few bins around a harmonic is highly destructive: it removes not only the target rotor but also overlapping speech formants, sidebands from other rotors, and any rotor energy that has leaked beyond the narrow suppression window. By the time we search for the third or fourth rotor, the spectrum is so mangled that the estimate is essentially noise.

PYIN is even worse off because we did not even attempt the greedy adaptation. It is a single- f_0 tracker; we just run it once per frame and replicate the single output four times. Of course it cannot resolve four independently varying rotors.

NMF is different: it is a true multi-pitch method. It jointly decomposes the spectrum into four harmonic combs without hard suppression. Yet it still fails, albeit by a smaller margin (MSE 615 vs 870–1380 for the greedy methods). Why? Overlapping harmonics from four rotors create ambiguity in the dictionary: many atoms have similar harmonic patterns, so the multiplicative updates spread energy across neighbouring candidates. Speech adds non-harmonic energy that NMF tries to explain with the same harmonic dictionary, creating spurious activations. And without a temporal model, the frame-wise factorisation can flip which activation is assigned to which rotor from one frame to the next. NMF improves on the greedy single-pitch hack, but the multi-pitch + speech problem is still too hard for a classical spectral decomposition with no learned temporal structure.

6 Sanity check: what happens without speech and with only one rotor?

So far the classical methods look useless. But remember they are single-pitch estimators, and DREGON-LM is a multi-pitch + speech mixture. Before writing them off completely, we checked how they behave on the original DREGON individual-motor recordings: clean recordings of a single rotor running at constant speed (no speech, no other rotors). We trimmed the start and end transients and evaluated the *single-pitch* versions of each method—no greedy suppression, no multi-rotor hack.

The results are summarised in Table 3 and Figure 5. PYIN and the matched filter are essentially perfect on most of these clips (MAE below 2 rev/s when they detect a periodicity). Even cepstral analysis and HPS, while noisier, stay within 5–10 rev/s of the target. NMF is mixed: on some motors it is accurate to within a few rev/s, but on others it locks onto the wrong harmonic or a spurious spectral peak and drifts far from the target. The single-pitch methods are not broken; they are simply solving the wrong problem on DREGON-LM.

Table 3: Mean MSE / MAE on clean single-rotor recordings (single-pitch methods, no greedy suppression). SimpleConv is shown as the best rotor and averaged over all four outputs.

Method	Mean MSE	Mean MAE
PYIN	240.8	6.2
Cepstral	742.7	15.1
HPS	1149.5	20.7
Matched Filter	168.2	6.9
NMF	599.7	11.5
SimpleConv (best rotor)	2027.3	41.1
SimpleConv (avg over 4)	2078.2	42.0

The really interesting result, however, is SimpleConv. On DREGON-LM it tracks all four rotor speeds quite accurately, so it is not simply memorising chunks of training audio; it generalises to unseen mixtures. On single-rotor recordings, though, it fails: all four channels predict roughly the same low value (10–50 rev/s instead of the true 50–90), so the four dotted lines in Figures 6 and 7 overlap perfectly. The network has clearly learned that there should be four rotors and that their speeds should differ and vary over time. When presented with a single constant rotor, it cannot reconcile that with its internal model of the problem, and all four channels collapse to the same confused estimate.

On the four-rotor synchronized recording (allMotors_70, Figure 8) SimpleConv performs much better (≈ 75 –83 rev/s vs target 70). The input now contains four rotors, which matches the network’s structural expectation; the absence of speech is apparently less important than the presence of the expected number of rotors.

7 Conclusion

On the DREGON-LM test set (multi-pitch + speech, low SNR) the classical baselines fail by two to three orders of magnitude versus a 1-M-parameter CNN. The four single-pitch methods (PYIN, cepstral, HPS, matched filter) are forced into a multi-pitch task with a crude greedy suppression hack that wipes out useful information. NMF, a true multi-pitch method, improves on the greedy approaches (MSE 615 vs 870–1380) but is still 86 $\times$ worse than SimpleConv and produces drifting, confused traces.

On clean single-rotor recordings the same classical methods are accurate to within a few rev/s. They are not broken; the multi-pitch task is simply harder than they were built for.

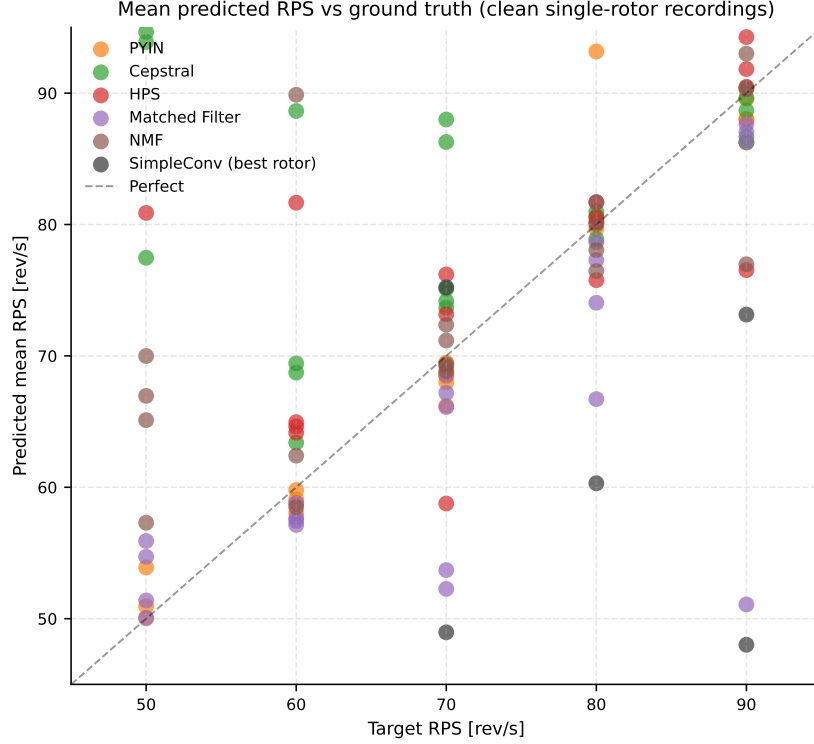


Figure 5: Mean predicted RPS vs target RPS on clean single-rotor recordings. Classical single-pitch methods cluster around the diagonal; SimpleConv predicts random low values because it has never seen a single-rotor input.

SimpleConv shows the reverse. It tracks the four rotor speeds accurately on DREGON-LM, so it is doing more than memorising training chunks; it generalises to new mixtures. Yet on single-rotor recordings all four channels collapse to the same wrong number (10–50 rev/s instead of 50–90). The fact that they all emit the same value is telling: the network has learned that there should be four rotors with different, varying speeds, and when that structure is absent it cannot produce a meaningful estimate. On the four-rotor recording (allMotors_70) it recovers speeds close to the target (≈ 75 –83 vs 70), because the input now matches the expected four-rotor structure.

Bottom line: even a proper multi-pitch classical method (NMF with harmonic priors) cannot solve this task reliably at low SNR. SimpleConv succeeds because it has learned the specific structure of the problem (four rotors, variable speeds, mixed with speech). It is not a general rotor-speed estimator, and its single-rotor failure shows it is relying on structural expectations rather than on independent per-rotor harmonic analysis.

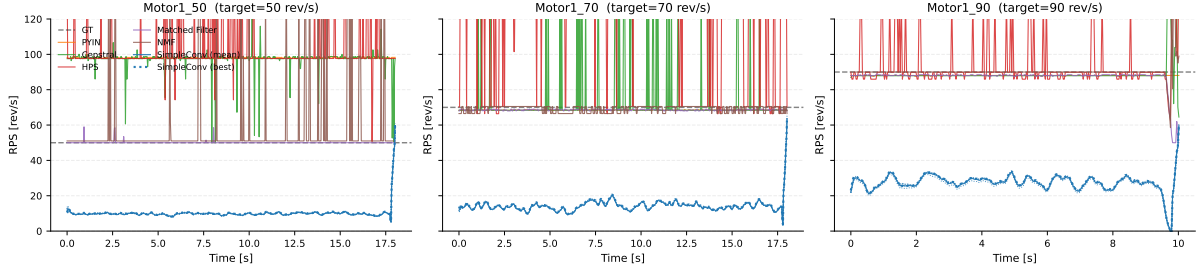


Figure 6: Predictions on clean single-rotor recordings (Motors 1 at 50, 70, 90 rev/s). Classical single-pitch methods track the constant target well. SimpleConv outputs four channels (dotted lines), but on single-rotor input all four collapse to the same wrong value, so the four dotted lines overlap perfectly. The solid line is their mean (identical to each channel), and one channel is shown bold to mark the “best” channel (they are all equal).

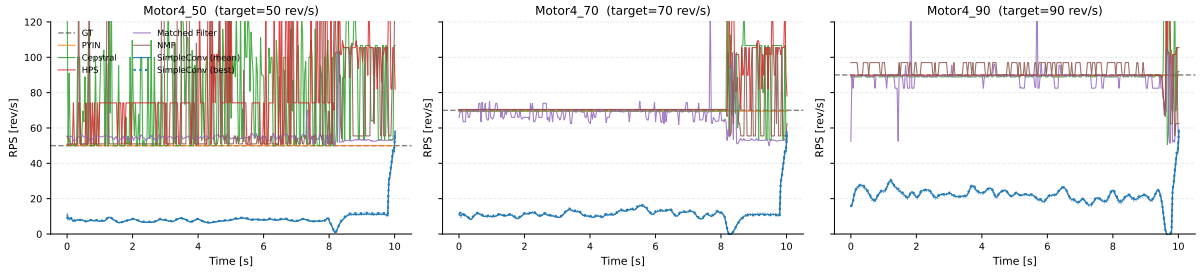


Figure 7: Predictions on clean single-rotor recordings (Motors 4 at 50, 70, 90 rev/s). Same pattern as Figure 6: all four SimpleConv channels output the same wrong value, so the dotted lines overlap.

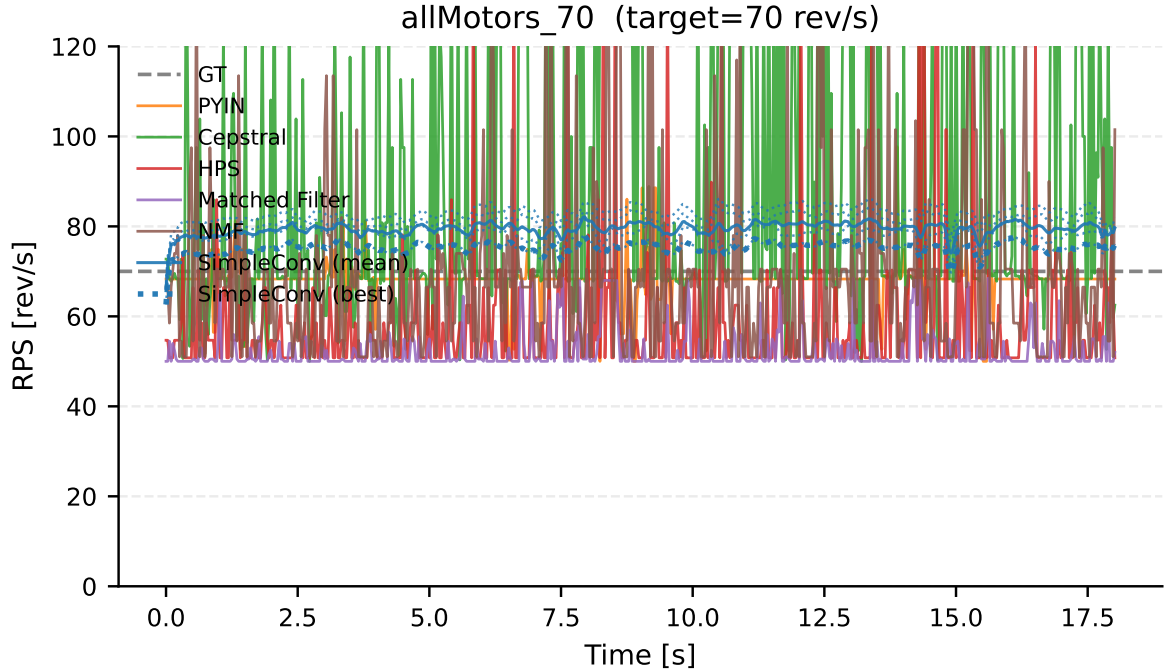


Figure 8: Predictions on the four-rotor synchronized recording (allMotors_70). Classical methods struggle with multiple overlapping combs. SimpleConv dotted = four channels, solid = mean, bold = best channel; the four outputs are now clustered near the true speed because the input matches training.